PH3204: Electronics Laboratory

Experiment 3

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1 Aim

In this experiment we aim to study the working of operational amplifiers (OP-AMP [LM741]) as inverting and non-inverting amplifier and its applications as adder and subtractor.

2 Theory

What are OP-AMPs?

The operational amplifier (OP AMP) is a direct coupled amplifier with a very high voltage gain, very high input impedance and very low output impedance. It can be used to perform a wide variety of linear and some nonlinear operations by merely changing a few external circuit elements. Originally OP AMP was designed to perform mathematical operations such as summation, subtraction, multiplication, differentiation, integration, etc in analog computer. That is why it is called *operational amplifier.*

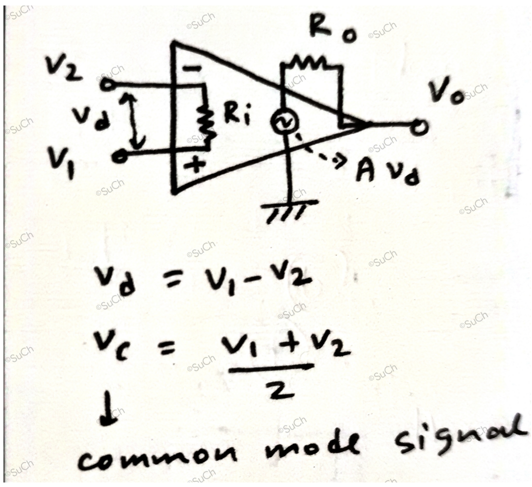

Figure 1: An A.C equivalent circuit of a practical OP AMP

Some characteristics of ideal OP-AMP include:

1. Open loop voltage gain is infinite ($A = \infty$).
2. Input impedance is infinite, $R_i = \infty$(so that any signal source can drive it without being loaded).

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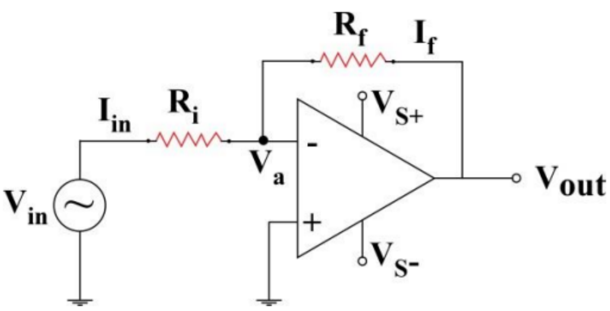
3. Output impedance is zero ($R_o = 0$) so that the output can drive any other device.
4. Bandwidth is infinite.
5. Perfect balance i.e output is zero and two input voltages are equal.
6. Characteristics do not drift with temperature.

We look at some of the important applications of OP-AMP that we shall perform in this experiment. Note that, throughout the experiment $V_S{}^{\pm}$ is kept fixed at ±15 V. The diagrams for inverting and non-inverting OP-AMP configurations were taken from the lab manual.

Inverting Amplifier



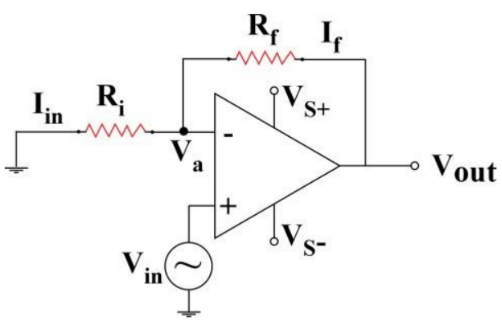
Now, in an OP AMP $V_{out} = Av_d$. Since for an ideal OP AMP $A \to \infty$, for a finite V_{out} we must have $v_d \to 0$. This implies $V_a = 0$ (*virtual ground*). Applying KCL gives

$$\therefore i = (V_{in} - 0)/R_i = (0 - V_{out})/R_f$$

$$\boxed{A_V = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}}$$

Where A_V is defined to be the **voltage gain**.

Non-inverting Amplifier

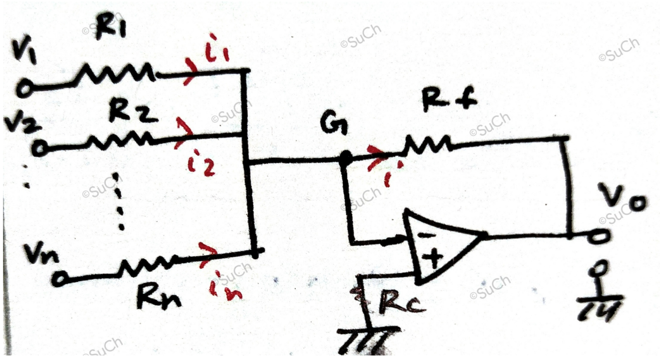


Similarly, proceeding for a finite V_{out} we must have $v_d = v_1 - v_2 \equiv V_{in}$ - $V_a = 0 \implies V_{in} = V_a$ (*virtual short*).

$$\therefore \frac{0 - V_a}{R_i} = \frac{V_a - V_{out}}{R_f}$$

$$\boxed{A_V = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}}$$

Adder



Applying the concept of virtual ground we have $V_G = 0$, thus

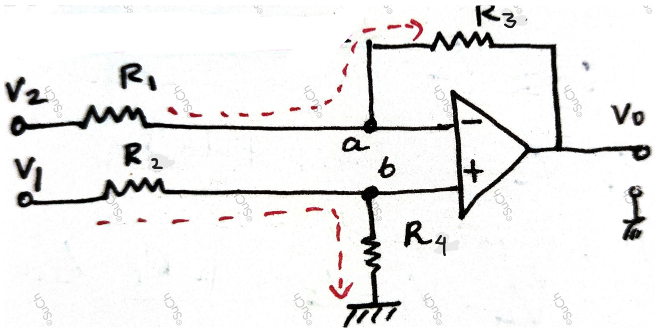
$$\sum_{k\leq n} i_k = i = -\frac{V_o}{R_f}$$

where $i_k = V_k/R_k$ is the individual branch currents.

$$\boxed{\therefore V_{out} = -R_f \sum_{k\leq n} \left(\frac{V_k}{R_k}\right)}$$

R_c is called compensating resistance, and it is chosen to be $R_C = R_f \parallel R_1 \parallel \ldots R_n$

Differential Amplifier



Applying the concept of virtual short, we have $V_a = V_b = x$ (say!).

$$\frac{V_2 - x}{R_1} = \frac{x - V_{out}}{R_3}$$

$$\frac{V_1 - x}{R_2} = \frac{x - 0}{R_4}$$

$$\Longrightarrow V_{out} = R_3\left(\frac{-V_2}{R_1} + \frac{V_1}{R_2}\left(\frac{1/R_3 + 1/R_1}{1/R_2 + 1/R_4}\right)\right)$$

In our experiment $R_1 \simeq R_2 \simeq R_3 \simeq R_4 = 1\text{k}\Omega$. Thus $\left(\frac{1/R_3+1/R_1}{1/R_2+1/R_4}\right) \approx 1$. So, final output voltage becomes

$$\boxed{\therefore V_{out} = R_3\left(\frac{V_1}{R_2} - \frac{V_2}{R_1}\right)}$$

Note that, if $R_1 = R_2 = R_3$, then it would be called **subtractor** circuit.

3 Data Analysis

Experimental gain is defined as $A_{exp} = V_{out} / V_{in}$, is reported as the average of all the readings of that particular configuration along with 1σ error. Theoretical gain needs to be calculated in first two cases. $\epsilon(\%)$ denotes the percentage deviation of experimentally observed value from the theoretical estimate.

3.1 Inverting Amplifier

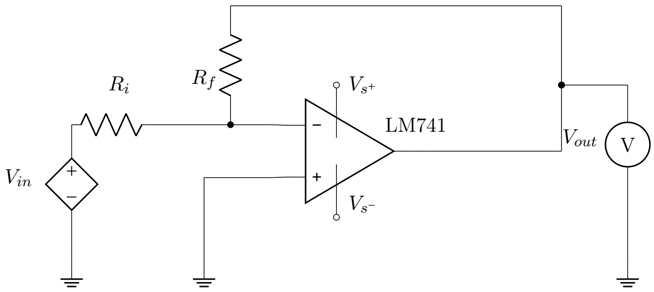


Figure 2: Inverting Amplifier Circuit diagram

$$A_{th} = -\frac{R_f}{R_i}$$

R_i (kΩ)	R_f (kΩ)	A_{th}	A_{exp}	$\epsilon(\%)$
1	2.2	-2.20	-2.12 ± 0.13	3.52
1	9.77	-9.77	-10.20 ± 0.79	4.40
1	22	-22	-21.28 ± 1.42	3.26
2.2	22	-10	-9.91 ± 0.12	0.87
2.2	10	-4.54	-4.49 ± 0.03	1.07

3.2 Non-inverting Amplifier

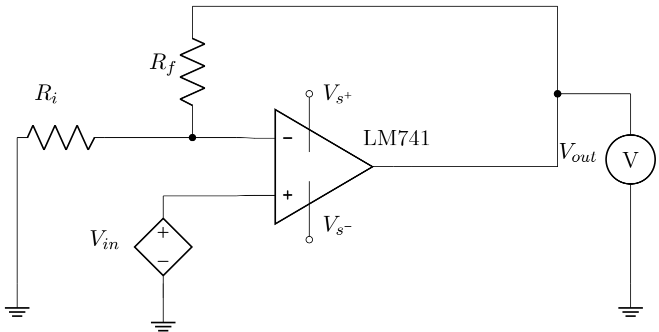


Figure 3: Non Inverting Amplifier circuit diagram

$$A_{th} = 1 + \frac{R_f}{R_i}$$

R_i (kΩ)	R_f (kΩ)	A_{th}	A_{exp}	$\epsilon(\%)$
2.2	10	5.54	5.48 ± 0.14	1.16
2.2	22	11	10.79 ± 0.46	1.90
1	22	23	23.33 ± 1.09	1.42
1	10	11	10.64 ± 0.87	3.25

3.3 Adder or Summing Amplifier

In our setup all the resistances were approximately equal so,

$$V_{out} \simeq -(V_1 + V_2 + V_3)$$

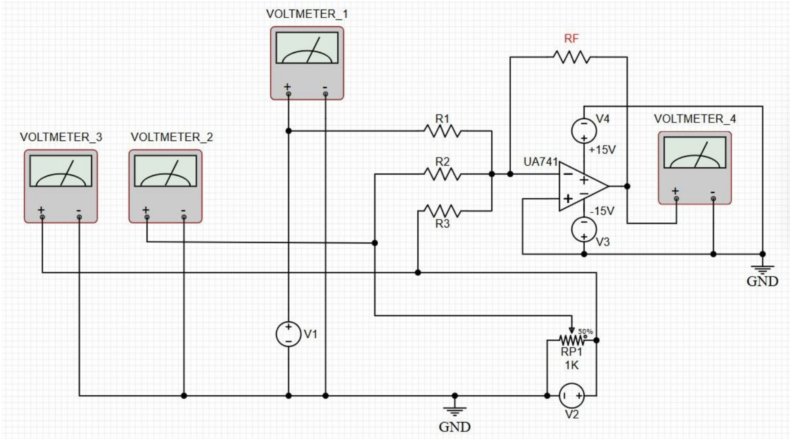


Figure 4: Adder circuit diagram

By varying V_1, V_2 and V_3 systematically we note (Table 10) the output voltage V_{out} as V_{obs} and compare it with theoretical estimate V_{th}. (Fig 5)

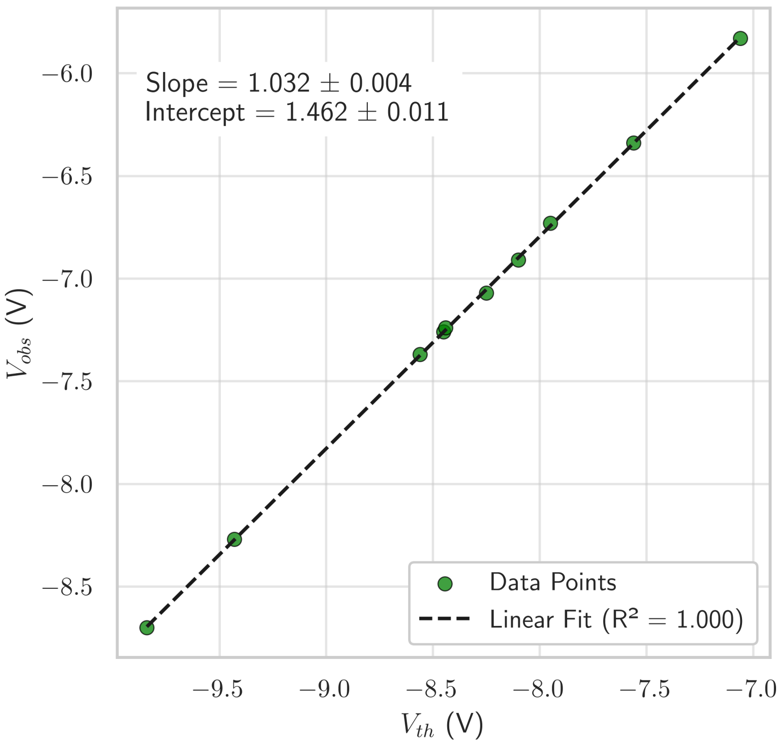


Figure 5: Comparison between V_{obs} and V_{th} in the adder circuit

In the above plot, a linear fit gives a slope of 1.032 ± 0.004 which very close to 1. This indicates a very strong correlation between our observed values and theoretical calculations.

3.4 Subtractor or Differential Amplifier

$$V_{out} = R_3\left(\frac{V_1}{R_2} - \frac{V_2}{R_1}\right)$$

In this setup we have, $R_1 = 1.005, R_2 = 1, R_3 = 0.999$ and $R_4 = 1.004$ kΩ respectively. By varying both V_1

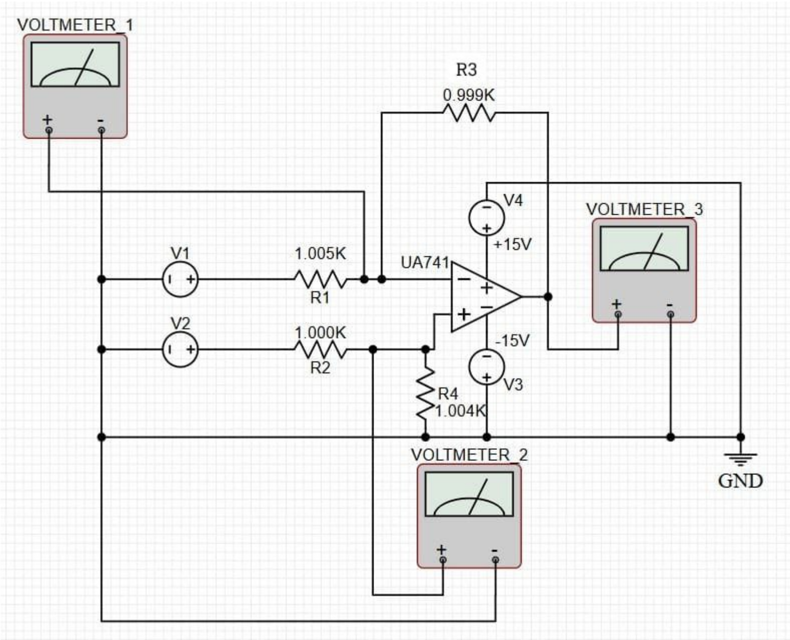


Figure 6: Subtractor circuit diagram

and V_2 systematically we note (Table 11) the output voltage V_{out} as V_{obs} and compare it with theoretical estimate V_{th}. (Fig 7)

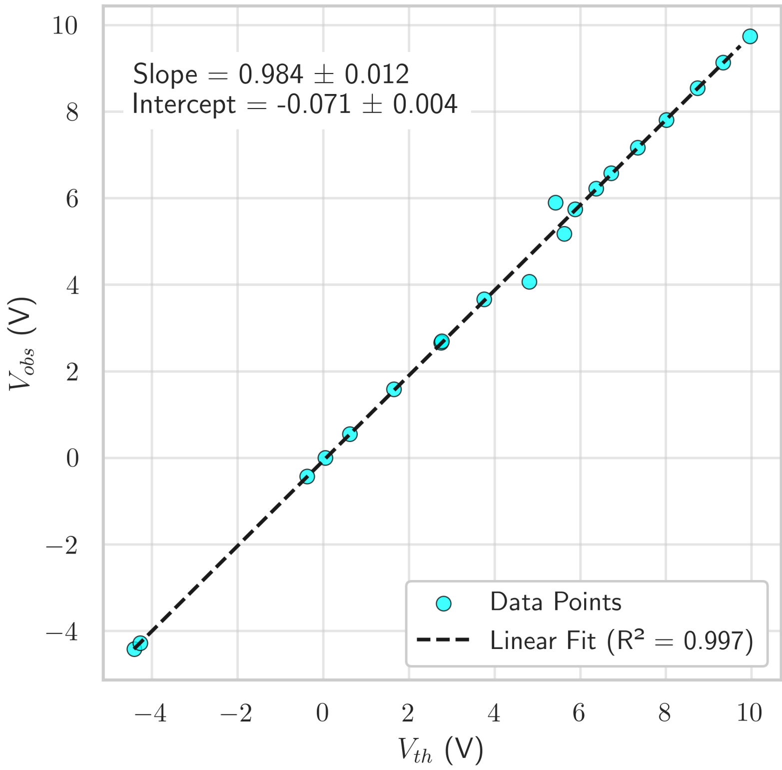


Figure 7: Comparison between V_{obs} and V_{th} in the subtractor circuit

In the above plot, a linear fit gives a slope of 0.984 ± 0.012 which very close to 1. This indicates a very strong correlation between our observed values and theoretical calculations.

4 Sources of Error

In this experiment, we note a few major sources of error:

- Loose connections or improper placement of wires on the breadboard could introduce additional resistance and voltage drops.
- The output of the given voltage source was not steady and was fluctuating.
- Our observation varies from theoretical expectations as our OP-AMPs were non ideal. We did not account for the internal and output impedances of the OP AMP . LM741 has small input offset voltage and bias current that may also cause errors.
- There might be a presence of instrumental error from measurements made in the multimeter due to improper calibration.
- We assumed all the resistances to be equal in the *adder* circuit, which in reality might not be so. This factor also contributed to it's deviation from theoretical value.
- The OP AMP were found to heat up, with increasing current, indicating it's properties could have drift with temperature.

5 Conclusion

This experiment successfully verifies the operation of OP AMP as inverting, non-inverting, summing and differential amplifier respectively within reasonable deviations from the theoretical expected values. These OP AMP can be further used to create complicated arithmetic circuits. This experiment can be improved further by compensating for the sources of error reported in the preceeding section.

References

- *B. Ghosh*, Fundamental Principles of Electronics 2021. ISBN: 9788187134398

Appendix

INVERTING AMPLIFIER

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
1	-2.19	-2.19	-2.20
1.5	-3.28	-2.19	-2.20
1.97	-4.3	-2.18	-2.20
2.48	-5.48	-2.21	-2.20
2.99	-6.53	-2.18	-2.20
3.5	-7.64	-2.18	-2.20
4.03	-8.79	-2.18	-2.20
4.54	-9.88	-2.18	-2.20
5.02	-10.91	-2.17	-2.20
5.48	-11.91	-2.17	-2.20
6.05	-12.49	-2.06	-2.20
6.5	-12.47	-1.92	-2.20
7.03	-12.45	-1.77	-2.20

Table 1: Data table for $R_i = 1$ kΩ , R_f = 2.2k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.02	-0.24	-12.00	-9.77
0.2	-1.98	-9.90	-9.77
0.42	-4.16	-9.90	-9.77
0.61	-6.04	-9.90	-9.77
0.8	-7.94	-9.93	-9.77
1.01	-9.99	-9.89	-9.77
1.2	-11.85	-9.88	-9.77

Table 2: Data table for $R_i = 1$ kΩ , R_f = 9.77k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.19	-1.84	-9.68	-10.00
0.41	-4.08	-9.95	-10.00
0.63	-6.25	-9.92	-10.00
0.8	-8	-10.00	-10.00
1.08	-10.74	-9.94	-10.00
1.19	-11.87	-9.97	-10.00

Table 3: Data table for $R_i = 2.2$ kΩ , R_f = 22k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.1	-2.17	-21.70	-22.00
0.2	-4.48	-22.40	-22.00
0.3	-6.62	-22.07	-22.00
0.4	-8.66	-21.65	-22.00
0.5	-10.82	-21.64	-22.00
0.6	-12.82	-21.37	-22.00
0.71	-12.89	-18.15	-22.00

Table 4: Data table for $R_i = 1$ kΩ , R_f = 22k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.27	-1.23	-4.56	-4.54
0.7	-3.13	-4.47	-4.54
1.03	-4.63	-4.50	-4.54
1.2	-5.4	-4.50	-4.54
1.53	-6.86	-4.48	-4.54
1.86	-8.32	-4.47	-4.54
2.06	-9.22	-4.48	-4.54
2.2	-9.88	-4.49	-4.54
2.36	-10.57	-4.48	-4.54

Table 5: Data table for $R_i = 2.2$ kΩ , R_f = 10k Ω

NON-INVERTING AMPLIFIER

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.10	0.57	5.70	5.54
0.20	1.12	5.60	5.54
0.31	1.72	5.55	5.54
0.42	2.32	5.52	5.54
0.52	2.87	5.52	5.54
0.70	3.85	5.50	5.54
0.90	4.98	5.53	5.54
1.21	6.70	5.54	5.54
1.50	8.28	5.52	5.54
1.80	9.92	5.51	5.54
2.01	11.04	5.49	5.54
2.10	11.53	5.49	5.54
2.20	12.07	5.49	5.54
2.41	13.20	5.48	5.54
2.50	13.69	5.48	5.54
2.32	12.71	5.48	5.54
2.60	14.18	5.45	5.54
2.76	14.17	5.13	5.54
2.80	14.17	5.06	5.54

Table 6: Data table for R_i = 2.2 kΩ , R_f = 10 k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.21	2.34	11.14	11.00
0.57	6.28	11.02	11.00
0.78	8.64	11.08	11.00
1.00	11.09	11.09	11.00
1.20	12.41	10.34	11.00
1.41	14.21	10.08	11.00

Table 7: Data table for R_i = 2.2 kΩ , R_f =22 k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.05	1.32	26.40	23.00
0.10	2.40	24.00	23.00
0.15	3.44	22.93	23.00
0.20	4.75	23.75	23.00
0.25	5.89	23.56	23.00
0.31	6.89	22.23	23.00
0.35	8.01	22.89	23.00
0.40	9.19	22.98	23.00
0.45	10.43	23.18	23.00
0.50	11.50	23.00	23.00
0.55	12.78	23.24	23.00
0.60	13.99	23.32	23.00
0.65	14.17	21.80	23.00

Table 8: Data table for R_i = 1 kΩ , R_f =22 k Ω

V_{in} (V)	V_{out} (V)	A_{exp}	A_{th}
0.20	2.25	11.25	11.00
0.38	4.21	11.08	11.00
0.59	6.64	11.25	11.00
0.80	8.74	10.93	11.00
1.03	11.24	10.91	11.00
1.20	13.15	10.96	11.00
1.40	14.09	10.06	11.00
1.62	14.09	8.70	11.00

Table 9: Data table for R_i =1 kΩ , R_f = 10 kΩ

ADDER

V_1 (V)	V_2 (V)	V_3 (V)	V_{obs} (V)	V_{th} (V)
3.09	3.08	2.08	-7.07	-8.25
3.55	2.5	2.4	-7.26	-8.45
3.56	2.5	2.04	-6.91	-8.1
3.56	2.5	1	-5.83	-7.06
3.56	2.5	1.5	-6.34	-7.56
3.56	2.5	2.5	-7.37	-8.56
4.08	2.86	2.9	-8.7	-9.84
4.08	2.86	2.49	-8.27	-9.43
4.08	2.86	1.5	-7.24	-8.44
4.08	2.86	1.01	-6.73	-7.95

Table 10: Data for Adder circuit

SUBTRACTOR

V_1 (V)	V_2 (V)	V_{obs} (V)	V_{th} (V)
4.44	0	-4.42	-4.44
4.44	4.47	0.00	0.01
4.44	5.04	0.55	0.57
4.44	6.07	1.58	1.60
4.44	7.17	2.66	2.69
4.44	8.18	3.66	3.70
4.44	9.23	4.07	4.74
4.44	10.05	5.18	5.55
6.04	11.44	5.90	5.34
6.04	12.74	6.58	6.63
4.44	0.14	-4.28	-4.30
4.44	4.04	-0.43	-0.42
4.44	7.19	2.69	2.71
5.26	11.12	5.74	5.80
5.71	12.06	6.22	6.28
6.58	13.89	7.17	7.23
7.17	15.15	7.81	7.90
7.83	16.54	8.54	8.62
8.38	17.69	9.13	9.21
8.93	18.87	9.74	9.84

Table 11: Data for subtractor circuit